

# **Structured Query Language (SQL)**

## **Introduction:**

SQL (Structured Query Language) is a programming language used to store, retrieve, manipulate, and manage data in relational databases.

Popular databases that use SQL include MySQL, PostgreSQL, Microsoft SQL Server, Oracle

## **General Database Structure**

### **1.Database Software:**

It can be regarded as common database software, such as MySQL,PostgreSQL, MySQL, etc.

### **2.Database:**

A **database** is an organized collection of data that can be easily stored, accessed, and managed.

### **3.Schema:**

This is a collection of Tables, which can be split and grouped according to logic, and some table details meta settings can be set on this layer, like a blueprint.

### **4.Table:**

The key section of the database. Tables record different data entities, such as users, products, etc.

### **5.Data:**

The most valuable part of the database is the data itself. Different types of data can be stored.

## Different types of MySQL

There are two main types in MySQL

- **DDL**(Data definition language)
- **DML**(Data Manipulation language)
- **TCL** (Transaction Control Language)

### Data definition language:

DDL (Data Definition Language) is used to create, modify, and delete the structure of database objects, such as tables, databases, and indexes.

|   | <b>Core commands</b> | <b>purpose</b>                              |
|---|----------------------|---------------------------------------------|
| 1 | Create               | makes new tables                            |
| 2 | Alter                | modifies table structures                   |
| 3 | Drop                 | deletes an entire table permanently)        |
| 4 | Truncate             | removes all records but keeps the structure |
| 5 | Rename               | changes the name of a database object       |

### Data Manipulation language:

DML (Data Manipulation Language) is used to insert, modify, and delete data inside database tables.

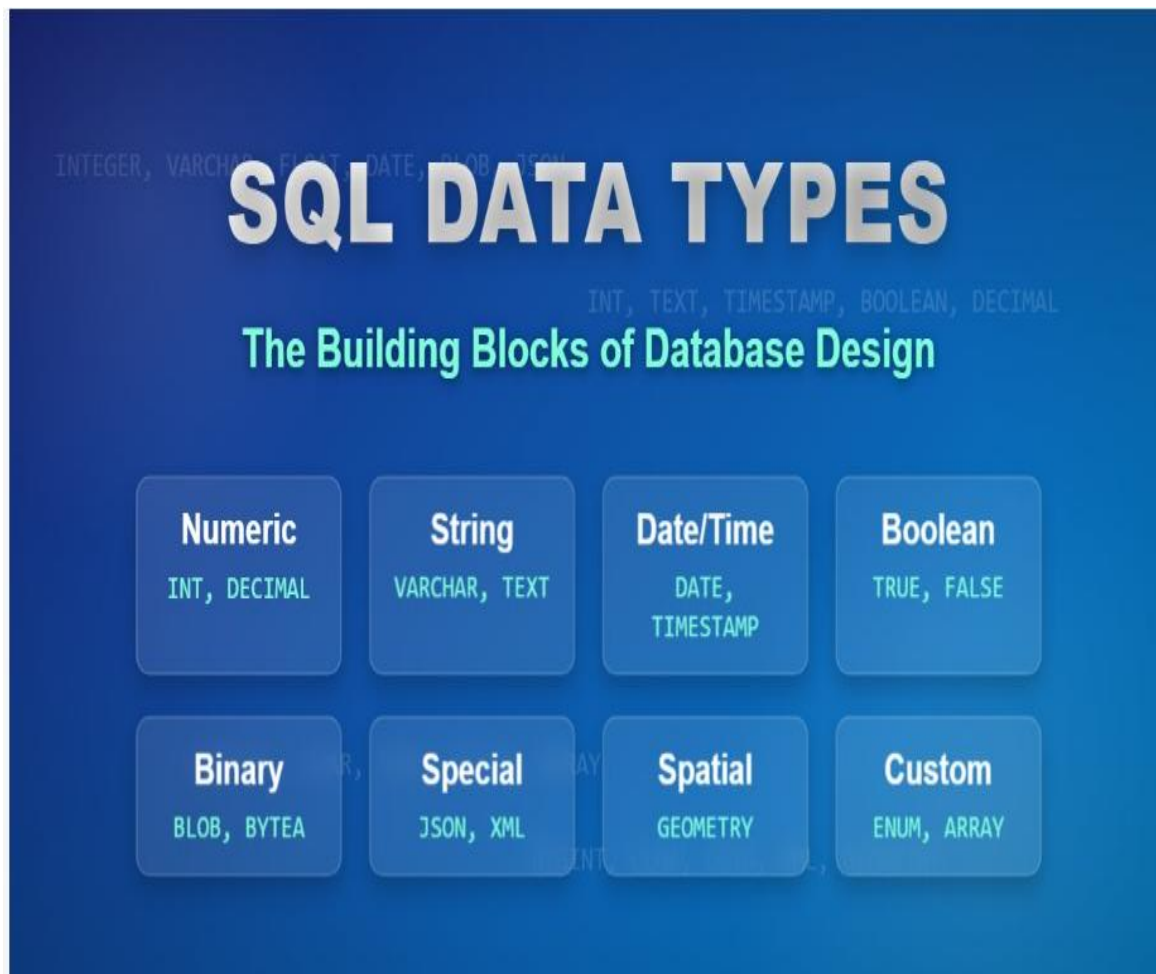
|   | <b>Core commands</b> | <b>purpose</b>           |
|---|----------------------|--------------------------|
| 1 | Insert               | adds new rows            |
| 2 | Update               | edits existing data      |
| 3 | Delete               | removes specific records |
| 4 | Select               | retrieves data           |

## Transaction Control Language:

**TCL (Transaction Control Language)** is used to **manage transactions in a database**. A transaction is a group of SQL operations treated as a single unit

|   | Core commands | purpose                                        |
|---|---------------|------------------------------------------------|
| 1 | Commit        | permanently saves the changes                  |
| 2 | Rollback      | undoes changes made in the current transaction |
| 3 | Savepoint     | creates a point within a transaction           |

## Datatypes of SQL



## Movie Database Management and Analytical Reporting Case Study:

#Create a database and use the database

```
create database Movies_DB;

use Movies_DB;

create table movie
(
rel_year int,
mov_name varchar(100),
act_name varchar(50),
lang varchar(30),
dir_name varchar(100)
);

INSERT INTO MOVIE VALUES
(1960,'Kalathur Kannamma','Kamal Haasan','Tamil','A. Bhimsingh'),
(1962,'Kannum Karalum','Kamal Haasan','Malayalam','K. S. Sethumadhavan'),
(1962,'Paadha Kaanikkai','Kamal Haasan','Tamil','K. Shankar'),
(1962,'Paarthaal Pasi Theerum','Kamal Haasan','Tamil','A. Bhimsingh'),
(1963,'Anandha Jodhi','Kamal Haasan','Tamil','V. N. Reddy'),
(1963,'Vanambadi','Kamal Haasan','Tamil','G. R. Nathan'),
(1970,'Maanavan','Kamal Haasan','Tamil','M. A. Thirumugam'),
(1971,'Annai Velankanni','Kamal Haasan','Tamil','Thankappan'),
(1971,'Anubhavangal Paalichakal','Mammootty','Malayalam','K. S. Sethumadhavan'),
(1972,'Kurathi Magan','Kamal Haasan','Tamil','K. S. Gopalakrishnan'),
(1973,'Arangetram','Kamal Haasan','Tamil','K. Balachander'),
(1973,'Kaalachakram','Mammootty','Malayalam','K. Narayanan'),
(1973,'Sollathaana Ninaikkiren','Kamal Haasan','Tamil','K. Balachander'),
(1974,'Anbu Thangai','Kamal Haasan','Tamil','S. P. Muthuraman'),
(1974,'Aval Oru Thodar Kathai','Kamal Haasan','Tamil','K. Balachander'),
(1974,'Gonasthavin Magal','Kamal Haasan','Tamil','A. P. Nagarajan'),
(1974,'Kanyakumari','Kamal Haasan','Malayalam','K. S. Sethumadhavan'),
(1974,'Naan Avenilalai','Kamal Haasan','Tamil','K. Balachander'),
(1974,'Panathukkaga','Kamal Haasan','Tamil','M. S. Senthil'),
(1974,'Paruva Kozham','Kamal Haasan','Tamil','Jose A.N. Fernando'),
(1974,'Vishnu Vijayam','Kamal Haasan','Malayalam','N. Sankaran Nair');
```

1.Retrieve all details from the movie table.

```
select * from MOVIES;
```

Solution:

| rel_year | mov_name                 | act_name     | lang      | dir_name             |
|----------|--------------------------|--------------|-----------|----------------------|
| 1960     | Kalathur Kannamma        | Kamal Haasan | Tamil     | A. Bhimsingh         |
| 1962     | Kannum Karalum           | Kamal Haasan | Malayalam | K. S. Sethumadhavan  |
| 1962     | Paadha Kaanikkai         | Kamal Haasan | Tamil     | K. Shankar           |
| 1962     | Paarthaal Pasi Theerum   | Kamal Haasan | Tamil     | A. Bhimsingh         |
| 1963     | Anandha Jodhi            | Kamal Haasan | Tamil     | V. N. Reddy          |
| 1963     | Vanambadi                | Kamal Haasan | Tamil     | G. R. Nathan         |
| 1970     | Maanavan                 | Kamal Haasan | Tamil     | M. A. Thirumugam     |
| 1971     | Annai Velankanni         | Kamal Haasan | Tamil     | Thankappan           |
| 1971     | Anubhavangal Paalichakal | Mammootty    | Malayalam | K. S. Sethumadhavan  |
| 1972     | Kurathi Magan            | Kamal Haasan | Tamil     | K. S. Gopalakrishnan |
| 1973     | Arangetram               | Kamal Haasan | Tamil     | K. Balachander       |
| 1973     | Kaalachakram             | Mammootty    | Malayalam | K. Narayanan         |
| 1973     | Sollathaana Ninaikkiren  | Kamal Haasan | Tamil     | K. Balachander       |
| 1974     | Anbu Thangai             | Kamal Haasan | Tamil     | S. P. Muthuraman     |
| 1974     | Aval Oru Thodar Kathai   | Kamal Haasan | Tamil     | K. Balachander       |

2.Display only movie names and actor names.

```
SELECT mov_name, act_name  
FROM movies;
```

Solution:

| mov_name                 | act_name     |
|--------------------------|--------------|
| Kalathur Kannamma        | Kamal Haasan |
| Kannum Karalum           | Kamal Haasan |
| Paadha Kaanikkai         | Kamal Haasan |
| Paarthaal Pasi Theerum   | Kamal Haasan |
| Anandha Jodhi            | Kamal Haasan |
| Vanambadi                | Kamal Haasan |
| Maanavan                 | Kamal Haasan |
| Annai Velankanni         | Kamal Haasan |
| Anubhavangal Paalichakal | Mammootty    |
| Kurathi Magan            | Kamal Haasan |
| Arangetram               | Kamal Haasan |
| Kaalachakram             | Mammootty    |
| Sollathaan Ninaikkiren   | Kamal Haasan |
| Anbu Thangai             | Kamal Haasan |
| Aval Oru Thodar Kathai   | Kamal Haasan |
| Gumasthavin Magal        | Kamal Haasan |
| Kanyakumari              | Kamal Haasan |

3.Display distinct movie languages.

```
select distinct lang from movies;
```

Solution:

| Result Grid | lang      |
|-------------|-----------|
| ►           | Tamil     |
|             | Malayalam |
|             | Telugu    |
|             | Kannada   |
|             | Hindi     |
|             | Bengali   |
|             | Silent    |
|             | English   |

4.Retrieve unique actor names.

```
SELECT DISTINCT act_name  
FROM movies;
```

Solution:



A screenshot of a database query result window. The window title is 'Query Results'. It shows a table with one column, 'act\_name'. The table contains 20 rows of actor names, each preceded by a small blue triangle icon. The names are: Kamal Haasan, Mammooty, Rajinikanth, Mohanlal, Vijay, Silambarasan, Ajith Kumar, Suriya, Trisha, Nayanthara, Karthi, Dhanush, Dulquer Salm..., Amala Paul, Rashmika Ma..., Manju Warior, Sai Pallavi, Parvathy Thi..., and Vijay Sethup...

| act_name        |
|-----------------|
| Kamal Haasan    |
| Mammooty        |
| Rajinikanth     |
| Mohanlal        |
| Vijay           |
| Silambarasan    |
| Ajith Kumar     |
| Suriya          |
| Trisha          |
| Nayanthara      |
| Karthi          |
| Dhanush         |
| Dulquer Salm... |
| Amala Paul      |
| Rashmika Ma...  |
| Manju Warior    |
| Sai Pallavi     |
| Parvathy Thi... |
| Vijay Sethup... |